

INDIGENE

ORGANIC CHEMISTRY

COMPLETE EXAM MASTER GUIDE — CONDITIONS, REAGENTS, REACTIONS,
ACIDITY, BASICITY, COLOUR TESTS

Prepared Exclusively For: **TOLA (INDIGENE)**

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100 Level | Nursing Science

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CHAPTER 1 — HYBRIDIZATION

Hybridization = mixing of atomic orbitals to form new equal-energy hybrid orbitals. Determines geometry and bond angles.

The Three Types

Type	Orbitals Mixed	Shape	Bond Angle	Example	Contains
sp ³	1s + 3p	Tetrahedral	109.5 deg	CH ₄ , C ₂ H ₆ , ROH	Single bonds only
sp ²	1s + 2p	Trigonal Planar	120 deg	C ₂ H ₄ , Benzene, C=O	One double bond
sp	1s + 1p	Linear	180 deg	C ₂ H ₂ , CO ₂ , HCN	One triple bond

Sigma vs Pi Bonds

Bond	Sigma	Pi	Rotation?	Why?
Single bond	1	0	FREE	No pi bond restricting it
Double bond	1	1	NO	Pi bond side-on overlap locks rotation
Triple bond	1	2	NO	Two pi bonds lock it completely

Key Rules

- Count GROUPS on the carbon. 4 groups = sp³. 3 groups (one double bond) = sp². 2 groups (one triple) = sp
- More s-character means SHORTER, STRONGER, and MORE ACIDIC bond: sp (50% s) > sp² (33% s) > sp³ (25% s)
- Pi bonds use unhybridized p orbitals. Sigma bonds use hybrid orbitals
- Pi bonds restrict rotation — this is WHY cis/trans (geometric) isomers exist
- Lone pairs count as groups for hybridization of N, O, S atoms (e.g. NH₃ = 4 groups = sp³)

THINGS TO NOTE: THINGS TO NOTE: (1) sp³ = tetrahedral, most common in organic chemistry. (2) sp² = flat — aldehydes, ketones, carboxylic acids, and benzene all have sp² carbons. (3) sp = linear — alkynes, CO₂, HCN. (4) Hybridization determines shape; shape determines reactivity. (5) In exams: draw structure first, then count groups on the target

carbon.

EXAM HACK: 4 things attached = sp^3 . 3 things (one double bond) = sp^2 . 2 things (one triple) = sp . DRAW the molecule, then count!

ALZHEIMER'S MODE: sp^3 = 4 arms spreading out in 3D like a diamond shape. sp^2 = flat triangle. sp = dead straight line. More s-character = stronger grip on electrons = shorter bond = more acidic C-H.

CHAPTER 2 — NOMENCLATURE AND CLASSES

IUPAC Naming — 5 Steps

- Step 1: Find the LONGEST continuous carbon chain — this is the parent chain
- Step 2: Number the chain from the end NEAREST to the first substituent or principal functional group
- Step 3: Name all substituents with their position numbers
- Step 4: List substituents ALPHABETICALLY — ignore di/tri/tetra when alphabetising
- Step 5: Add the correct suffix based on the HIGHEST priority functional group

Parent Chain Prefixes — MEMORISE

Carbons	Prefix	Carbons	Prefix	Carbons	Prefix
1	Meth-	4	But-	7	Hept-
2	Eth-	5	Pent-	8	Oct-
3	Prop-	6	Hex-	9 / 10	Non- / Dec-

Suffix Priority Order

Priority	Functional Group	Suffix	Example
1 Highest	Carboxylic acid -COOH	-oic acid	Ethanoic acid
2	Ester -COOR	-oate	Methyl ethanoate
3	Amide -CONH ₂	-amide	Ethanamide
4	Aldehyde -CHO	-al	Ethanal
5	Ketone -CO-	-one	Propanone
6	Alcohol -OH	-ol	Ethanol
7	Amine -NH ₂	-amine	Methanamine
8 Lowest	Alkene/Alkyne	-ene / -yne	Propene / Propyne

THINGS TO NOTE: THINGS TO NOTE: (1) Longest chain wins. (2) Give substituents the **LOWEST** possible set of numbers. (3) Cyclic compounds: add cyclo- prefix. (4) Alphabetise by the substituent name, ignoring multiplying prefixes (di, tri, tetra). (5) When acid **AND** alcohol are present, carboxylic acid suffix takes priority.

EXAM HACK: Mnemonic: My Enormous Pig Brought Home Old Nasty Dead Crabs = Meth Eth Prop But Pent Hex Hept Oct Non Dec

ALZHEIMER'S MODE: Naming = giving a molecule an address. Find the longest road (parent chain). Number from the end closest to a branch. Name every branch with its number. Alphabetise. Add the suffix at the end. Done.

CHAPTER 3 — HOMOLOGOUS SERIES AND FUNCTIONAL GROUPS

A homologous series = family of compounds with the same functional group and same general formula, differing by $\text{-CH}_2\text{-}$ (14 mass units). Properties change GRADUALLY along the series.

Complete Functional Group Table

Functional Group	Structure	Class	General Formula	Suffix	BP Rank
None (alkyl)	C-C single bonds only	Alkane	$\text{C}_n\text{H}_{2n+2}$	-ane	Lowest
C=C double bond	Two carbons double bonded	Alkene	C_nH_{2n}	-ene	Low
C triple C	Two carbons triple bonded	Alkyne	$\text{C}_n\text{H}_{2n-2}$	-yne	Low
-OH hydroxyl	Single O bonded to C and H	Alcohol	$\text{C}_n\text{H}_{2n+1}\text{OH}$	-ol	High — H-bonding
-CHO carbonyl at end	C=O at chain terminus	Aldehyde	RCHO	-al	Moderate
-CO- carbonyl in chain	C=O in chain middle	Ketone	RCOR	-one	Moderate
-COOH carboxyl	C=O and OH on same C	Carboxylic acid	RCOOH	-oic acid	Highest
-NH ₂ amino	N bonded to C and 2 H	Amine	RNH ₂	-amine	Moderate-high
-O- ether linkage	O between two C groups	Ether	ROR	ether or oxy-	Low
-COO- ester linkage	C=O-O between two C groups	Ester	RCOOR	-oate	Moderate
-CONH ₂ amide	C=O bonded to N	Amide	RCONH ₂	-amide	High
-X halide	C bonded to F Cl Br or I	Haloalkane	RX	halo-	Low-moderate

Physical Property Trends

Property	Trend	Reason	Example Comparison
Boiling point	RISES with chain length	More surface area — stronger London dispersion forces	Methane -162 degrees C vs Hexane 69 degrees C
Boiling point	FALLS with more branching	Less surface area between molecules — weaker forces	n-butane -0.5 vs isobutane -11.7 degrees C
Boiling point	RISES with -OH or -NH ₂ groups	Strong intermolecular hydrogen bonding	Ethanol 78 degrees vs ethane -89 degrees C (same C count)
Solubility in water	FALLS with chain length	Hydrophobic tail dominates over polar -OH	Methanol: miscible. Hexanol: slightly soluble only
Melting point	Generally RISES with chain length	More ordered crystal packing in solid state	Higher MW = more organised lattice = higher MP

THINGS TO NOTE: THINGS TO NOTE: (1) H-bonding requires H bonded to F, O, or N on one molecule and a lone pair of F, O, or N on another. (2) Branching **LOWERS** BP by reducing surface contact area. (3) Amides have very high BP because -NH₂ gives **TWO** H-bond donors. (4) Aldehydes and ketones have **NO** H-bond donors (no OH/NH) but **ARE** polar due to C=O — moderate BP. (5) Ether BP is similar to alkane of same MW — no H-bonding between ether molecules.

EXAM HACK: BP order top to bottom: RCOOH > ROH > RCONH₂ > RNH₂ > RCHO/RCOR > RCOOR > ROR > RX > RH. More H-bonding = higher BP. Always!

ALZHEIMER'S MODE: Same family = same personality (same reactions), different sizes. OH groups make molecules clingy (H-bonds). More clingy = harder to pull apart = higher boiling point. Branching makes molecules round and smooth — less surface area = less clingy = lower BP.

CHAPTER 4 — ORGANIC REACTIONS

Main Reaction Types — Conditions, Reagents, Products

Type	What Happens	Reagents and Conditions	Products	Things to Note
Addition	Two molecules combine into ONE product; nothing is lost	H ₂ /Ni heat (hydrogenation); HX no peroxide (Markovnikov); HBr + peroxide (anti-Markovnikov); X ₂ /CCl ₄ ; H ₂ O/H ⁺ (hydration of alkene); HOX/H ₂ O (halohydrin)	Saturated product — alkane, alcohol, dihalide, halohydrin	ONLY alkenes and alkynes undergo addition. Which rule applies depends on whether peroxide is present.
Free radical substitution	H on alkane replaced by halogen; chain mechanism: initiation, propagation, termination	X ₂ (Cl ₂ or Br ₂) + UV light (hν). No catalyst needed. Alkane substrate	Haloalkane + HX	UV light (hν) is ESSENTIAL to generate free radicals. Mixture of mono/di/tri-substituted products formed. Less selective than ionic mechanisms.
Nucleophilic substitution	Nu replaces leaving group on haloalkane. Either S _N 1 or S _N 2 depending on conditions (see separate table)	Nucleophile: NaOH (gives alcohol), KCN (gives nitrile), NH ₃ (gives amine), NaOR (gives ether), AgNO ₃ (gives alcohol + AgX). Conditions: S _N 1 = polar protic solvent + heat; S _N 2 = polar aprotic + strong Nu	Depends on nucleophile used	Leaving group ability: I > Br > Cl > F. The better the leaving group, the faster the reaction. F is poor LG despite high electronegativity.
Electrophilic addition to benzene (EAS)	Electrophile substitutes H on benzene ring; aromaticity preserved	See full EAS table in Chapter 5. Lewis acid catalyst essential for halogenation and Friedel-Crafts	Substituted benzene	Preferred over addition because addition would destroy aromaticity (lose ~150 kJ/mol stabilization). Directing groups control where next group goes.

Type	What Happens	Reagents and Conditions	Products	Things to Note
Elimination	Two atoms/groups removed from adjacent carbons; double bond forms	Strong base: KOH or NaOH in ETHANOL (not water — water gives SN2 instead); heat required. Dehydration: conc. H ₂ SO ₄ at 170 degrees C	Alkene + HX (or H ₂ O)	Zaitsev rule: major product = most substituted alkene. Competing with SN2 — heat favours elimination. Use ethanol solvent to favour E2 over SN2.
Oxidation	Oxygen added or hydrogen removed; carbon oxidation state increases	Mild: PCC or K ₂ Cr ₂ O ₇ /H ₂ SO ₄ controlled. Strong: KMnO ₄ /H ⁺ or excess K ₂ Cr ₂ O ₇ /H ₂ SO ₄ . Very strong: O ₃ (ozonolysis)	1 degree alcohol to aldehyde to carboxylic acid. 2 degree alcohol to ketone. Alkene to diol or cleaved products	1 degree: mild = aldehyde; excess = acid. 2 degree = ketone (stops). 3 degree = no reaction. Dichromate turns orange to green. KMnO ₄ turns purple to colourless then brown.
Reduction	Hydrogen added or oxygen removed; carbon oxidation state decreases	H ₂ /Ni (catalytic hydrogenation of C=C and C=O). NaBH ₄ (mild — reduces aldehydes/ketones ONLY). LiAlH ₄ (strong — reduces all C=O, COOH, COOR, COCl, CONHR). Zn-Hg/HCl or NH ₂ NH ₂ /KOH (C=O to CH ₂)	Alkene to alkane; Aldehyde to 1 degree alcohol; Ketone to 2 degree alcohol; Acid to 1 degree alcohol	NaBH ₄ cannot reduce COOH or COOR. LiAlH ₄ reduces everything. Both require dry ether solvent. Clemmensen (Zn-Hg/HCl) = acidic conditions. Wolff-Kishner (NH ₂ NH ₂ /KOH) = basic conditions.
Condensation	Two molecules join with expulsion of a small molecule (usually H ₂ O or HCl)	RCOOH + ROH with H ₂ SO ₄ cat (ester); amino acid + amino acid (peptide bond); acyl chloride + amine (amide)	Ester + H ₂ O; Peptide + H ₂ O; Amide + HCl	Always identify the small molecule expelled. H ₂ O is the most common. HCl is expelled in reactions of acyl chlorides with amines or alcohols.

Type	What Happens	Reagents and Conditions	Products	Things to Note
Hydrolysis	Large molecule split apart by reaction with water; reverse of condensation	H ₂ O + H ⁺ (acid catalysis) OR H ₂ O + OH ⁻ (base catalysis); heat usually required	Ester to carboxylic acid + alcohol; Amide to acid + amine; Protein to amino acids; Starch to glucose; Nitrile to amide to acid	Base hydrolysis of esters (saponification) is IRREVERSIBLE — gives carboxylate SALT. Acid hydrolysis of esters is reversible.

SN1 vs SN2 — Full Comparison

Feature	SN1	SN2
Full name	Substitution Nucleophilic Unimolecular	Substitution Nucleophilic Bimolecular
Steps	2 steps: slow then fast	1 step: concerted (all at once)
Rate law	Rate = k[substrate] only	Rate = k[substrate][nucleophile] — both affect speed
Intermediate formed	Carbocation (planar, sp ² carbon)	No intermediate — single transition state
Stereochemistry	Racemisation (attack from both faces of flat carbocation)	Walden inversion (nucleophile attacks from BACK — like umbrella flipping)
Best substrate	Tertiary (3 degree) — stable carbocation; also 2 degree	Primary (1 degree) — no steric hindrance at back; methyl is fastest
Nucleophile strength needed	Weak nucleophile OK — even water works	Strong nucleophile ESSENTIAL: OH ⁻ , CN ⁻ , I ⁻ , Br ⁻ , RS ⁻
Solvent	Polar PROTIC: water, alcohols (ROH) — stabilise carbocation by solvation	Polar APROTIC: DMSO, DMF, acetone — don't solvate nucleophile, keeps it reactive
Rearrangements?	YES — hydride or methyl shifts possible before attack	NO — concerted mechanism means no time for rearrangement
Temperature	Heat FAVOURS SN1 (and elimination)	Room temperature or lower favours SN2

Feature	SN1	SN2
Example	(CH ₃) ₃ CCl + H ₂ O at heat — tert-butanol	CH ₃ Br + NaOH at room temp — methanol

Markovnikov vs Anti-Markovnikov

Rule	Condition	Mechanism	H goes to...	X goes to...	Product Example
Markovnikov	HX addition — NO peroxide present	Ionic via carbocation: more stable = more substituted carbocation favoured	Carbon with MORE H's (less substituted C)	Carbon with FEWER H's (more substituted C)	CH ₃ CH=CH ₂ + HBr gives CH ₃ CHBrCH ₃ (2-bromopropane)
Anti-Markovnikov	HBr + peroxide ROOR only	Free radical chain: radical adds to give more stable (more substituted) radical	Carbon with FEWER H's (less substituted C)	Carbon with MORE H's (more substituted C)	CH ₃ CH=CH ₂ + HBr/ROOR gives CH ₃ CH ₂ CH ₂ Br (1-bromopropane)

THINGS TO NOTE: THINGS TO NOTE: (1) Anti-Markovnikov ONLY works with HBr + peroxide — NOT HCl or HI. (2) Zaitsev rule for elimination: most substituted alkene is major product. Hofmann product = least substituted (formed with bulky bases). (3) Leaving group ability order: I > Br > Cl > F — iodide is the best leaving group for substitution. (4) SN2 CANNOT occur at tertiary carbons (too much steric hindrance around the back). SN1 CANNOT occur at primary carbons (primary carbocation too unstable). (5) Competing reactions: SN1 vs E1 (both favoured by heat + polar protic). SN2 vs E2 (SN2 at RT, E2 with heat + strong base in ethanol).

EXAM HACK: SN1 = tertiary + weak Nu + polar protic + heat. SN2 = primary + strong Nu + polar aprotic + RT. Markovnikov = H goes to crowded side. Peroxide + HBr = anti-Markovnikov.

ALZHEIMER'S MODE: SN1 = two steps: first carbocation forms (slow), then Nu attacks from both faces (racemisation). SN2 = one smooth backside attack like an umbrella flipping inside out (inversion). Markovnikov: H is greedy and joins the side already crowded with H atoms.

CHAPTER 5 — BENZENE AND AROMATIC CHEMISTRY

Benzene (C₆H₆): flat hexagonal ring, all C-C bonds EQUAL at 1.40 angstroms due to resonance. Prefers SUBSTITUTION over addition to preserve aromaticity and its 150 kJ/mol stabilization energy.

Properties

Property	Value	Exam Significance
Molecular formula	C ₆ H ₆	Degree of unsaturation = 4 (3 double bonds + 1 ring)
C-C bond length	1.40 Å (between single 1.54 Å and double 1.34 Å)	Proves full electron delocalization — not alternating single/double
Bond angles	120 degrees	All 6 carbons are sp ² hybridized
Pi electrons	6 (Huckel: 4n+2, n=1, gives 6)	Satisfies aromaticity — highly stable
Melting point	5.5 degrees C	Liquid at slightly above room temperature
Boiling point	80.1 degrees C	Low for a 6-carbon aromatic
Solubility	Insoluble in water	Nonpolar molecule — dissolves in organic solvents
Combustion	Burns with sooty smoky flame	High C:H ratio (6:6)
Preferred reaction	EAS — substitution	Addition would destroy aromaticity
Toxicity	Toxic and carcinogenic	Handle with extreme care in the lab

All EAS Reactions — Conditions, Reagents, Products, Notes

Reaction	Reagents	Conditions	Electrophile	Product	Things to Note
Nitration	Conc. HNO ₃ + Conc. H ₂ SO ₄	50 to 55 degrees C — must control temperature carefully	NO ₂ ⁺ (nitronium ion)	Nitrobenzene	H ₂ SO ₄ protonates HNO ₃ to generate NO ₂ ⁺ . Above 55 degrees C gives dinitration. H ₂ SO ₄ is both catalyst and dehydrating agent.

Reaction	Reagents	Conditions	Electrophile	Product	Things to Note
Sulfonation	Fuming H ₂ SO ₄ (oleum, which is SO ₃ dissolved in H ₂ SO ₄)	Heat around 60 degrees C	SO ₃ (electrophilic sulfur)	Benzenesulfonic acid	REVERSIBLE — SO ₃ H group can be removed by steam (dilute H ₂ SO ₄ + heat). Used to temporarily BLOCK a ring position.
Halogenation	Cl ₂ or Br ₂ + Lewis acid (AlCl ₃ for Cl ₂ ; FeBr ₃ for Br ₂)	Room temperature, strictly anhydrous	Cl ⁺ or Br ⁺ (activated complex with Lewis acid)	Chlorobenzene or Bromobenzene	Lewis acid activates X ₂ by polarising X-X bond. MUST be anhydrous — water destroys Lewis acid. F ₂ = too reactive (no catalyst needed). I ₂ = too unreactive (needs oxidising agent).
Friedel-Crafts Alkylation	Alkyl halide RX + AlCl ₃	Anhydrous, gentle heat	Carbocation R ⁺ or R-AlCl ₃ complex	Alkylbenzene (e.g. toluene, ethylbenzene)	PROBLEMS: (1) Poly-alkylation — product is more reactive than starting material. (2) Carbocation rearrangements possible. Use acylation (below) to avoid these problems.
Friedel-Crafts Acylation	Acyl chloride RCOCl + AlCl ₃ (or RCOOH + P ₂ O ₅)	Anhydrous, gentle heat	Acylium ion RCO ⁺	Aryl ketone (e.g. acetophenone)	NO rearrangement — acylium is resonance-stabilised. Product is LESS reactive than benzene (C=O is EWG and meta director) — reaction stops at mono-substitution. Then reduce C=O with Clemmensen to get pure alkylbenzene.

Directing Groups — Where Next Substituent Goes

Type	Examples	Position Directed	Effect on Rate	Mechanism of Directing
Activating Ortho/Para directors	Strongly activating: -OH, -OR (ethers), -NH ₂ , -NHR, -NR ₂ , -NHCOR. Moderately: -R (alkyl), -Ar (aryl)	Ortho (1,2) AND Para (1,4)	Activating — SPEEDS UP EAS significantly	These groups donate electrons to the ring by resonance and/or induction. The ortho and para positions become electron-rich and ATTRACT the incoming electrophile. Ortho/para products are lower energy transition states.

Type	Examples	Position Directed	Effect on Rate	Mechanism of Directing
Deactivating Ortho/Para directors	Halogens only: -F, -Cl, -Br, -I	Ortho (1,2) AND Para (1,4)	Deactivating — slows EAS but still gives ortho/para	SPECIAL CASE — EXAM FAVOURITE: Halogens withdraw electrons by INDUCTION (sigma bond, deactivating) but DONATE electrons by RESONANCE (lone pairs into ring, directing ortho/para). Net = deactivating BUT ortho/para directing.
Deactivating Meta directors	-NO ₂ , -CN, -COOH, -CHO, -COR, -SO ₃ H, -CF ₃ , -NR ₃ ⁺	Meta (1,3) ONLY	Strongly deactivating — greatly SLOWS EAS	These groups withdraw electrons from the ring by both resonance AND induction. The ortho and para positions become very electron-poor (destabilised transition states). Meta is relatively less poor — electrophile goes there by default.

THINGS TO NOTE: THINGS TO NOTE: (1) EAS mechanism: electrophile attacks ring to form Wheland intermediate (arenium ion, sp³ carbon formed) then H⁺ leaves to restore aromaticity. (2) With TWO groups on the ring: if they direct to the SAME position, that is where reaction occurs. If they conflict, the stronger activating group wins. (3) Friedel-Crafts fails completely if ring has a strongly deactivating meta director like -NO₂. (4) Aniline (-NH₂) complexes with AlCl₃ in Friedel-Crafts — must protect as acetamide (-NHCOCH₃) first. (5) For di-substituted benzenes: use ortho (1,2), meta (1,3), para (1,4) — or E/Z naming for the substituents.

EXAM HACK: Electron DONOR = activating = ortho/para director. Electron WITHDRAWER = deactivating = meta director. EXCEPTION: Halogens are deactivating BUT ortho/para directors. This is the most tested benzene fact — know it cold!

ALZHEIMER'S MODE: Activating groups = turning up the music at a party. Electrophiles crowd around at spots 1,2 and 1,4 (ortho and para). Meta directors = turning the music off. Only the hardest electrophiles show up and they stand at 1,3 (meta). Halogens = weird guests who quiet the party but still point people to ortho/para spots.

CHAPTER 6 — ISOMERISM

Isomers = same molecular formula, different structural or spatial arrangement.

Complete Classification

Type	Subtype	Definition	What You Need	Example
Structural	Chain isomers	Different carbon skeleton arrangement	Same formula, different chain branching	n-butane vs isobutane (both C ₄ H ₁₀)
Structural	Position isomers	Same chain, different position of functional group	Same chain and group, different location number	1-propanol vs 2-propanol
Structural	Functional group isomers	Same formula, completely different functional group	Same molecular formula only	Ethanol vs dimethyl ether (both C ₂ H ₆ O)
Stereo	Geometric (cis/trans)	Same connectivity; groups same or opposite sides of C=C	Need C=C PLUS 2 different groups on EACH carbon of the double bond	cis-2-butene vs trans-2-butene
Stereo	Optical (enantiomers)	Non-superimposable mirror images — rotate polarised light opposite directions	Chiral center: one carbon with 4 DIFFERENT groups	D-alanine vs L-alanine
Stereo	Diastereomers	Stereoisomers that are NOT mirror images	Multiple chiral centers present	Meso-tartaric acid; cis and trans-1,2-dimethylcyclohexane

Chirality and Optical Activity

- Chiral center: carbon bonded to 4 COMPLETELY DIFFERENT groups — marked with asterisk (*)
- Enantiomers rotate plane-polarized light: (+) dextrorotatory (clockwise); (-) levorotatory (anticlockwise)
- Racemic mixture: 50:50 enantiomers — net rotation = 0 (optically INACTIVE despite chiral centers)
- Meso compound: has chiral centers but internal plane of symmetry makes it optically INACTIVE
- Maximum stereoisomers = 2 to the power n, where n = number of chiral centers, then subtract meso forms

R/S Configuration (CIP Rules)

- Step 1: Rank 4 groups by atomic number: highest atomic number = priority 1, lowest = priority 4
- Step 2: If tie at first atom, go to next atom along the chain to break the tie
- Step 3: Orient molecule so that priority 4 (lowest) points AWAY from you into the page
- Step 4: Trace 1 to 2 to 3: CLOCKWISE = R (Rectus) | ANTICLOCKWISE = S (Sinister)

Concept	Definition	Key Exam Fact
Chiral center (asymmetric C)	Carbon with 4 different groups attached	Look for C bearing H, OH, R, and R-prime all different
Enantiomers	Non-superimposable mirror images	Same physical properties EXCEPT optical rotation. $[\alpha]$ equal and opposite.
Diastereomers	Stereoisomers that are NOT mirror images	DIFFERENT physical and chemical properties — different melting points, BPs
Meso compound	Has chiral centers but internal mirror plane cancels activity	Optically inactive despite having chiral centers
Racemic mixture	Equal parts (+) and (-) enantiomers	Optically inactive — rotations exactly cancel each other out
E/Z nomenclature	Used when cis/trans is ambiguous (three different groups)	Z = same side (zusammen = together); E = opposite (entgegen = opposite)

THINGS TO NOTE: THINGS TO NOTE: (1) Geometric isomers CANNOT interconvert at room temperature — rotation around C=C is restricted. (2) Conformational isomers (like staggered vs eclipsed ethane) CAN interconvert freely — they are NOT true isomers. (3) Fischer projections: horizontal bonds point TOWARD you; vertical bonds point AWAY from you. (4) Specific rotation $[\alpha] = \text{observed rotation} / (\text{concentration} \times \text{path length in dm})$. (5) Number of stereoisomers = 2^n maximum. Meso forms reduce this count.

EXAM HACK: Geometric: need C=C + different groups on each end. Optical: need 4 different groups on one carbon. R = clockwise. S = anticlockwise. Max stereoisomers = 2 to the power n.

ALZHEIMER'S MODE: Chiral carbon = your two hands — look identical but cannot be stacked. Cis = same side (same city). Trans = opposite sides (transcontinental = crossing over). Meso = molecule that cancels itself out — like having a mirror inside it.

CHAPTER 7 — ALCOHOLS

Alcohols contain the -OH (hydroxyl) group bonded to a saturated carbon. Formula ROH. High boiling points, versatile reactivity.

Classification and Acidity

Class	Structure	C bearing -OH attached to...	Acidity	Example
Primary (1 degree)	RCH ₂ -OH	1 other carbon	Most acidic aliphatic alcohol	Ethanol CH ₃ CH ₂ OH (pK _a 16)
Secondary (2 degree)	R ₂ CH-OH	2 other carbons	Less acidic	2-propanol (pK _a 17)
Tertiary (3 degree)	R ₃ C-OH	3 other carbons	Least acidic	2-methyl-2-propanol (pK _a 19)

Acidity Factor	Effect	Reason	Example
More alkyl groups (EDG)	DECREASE acidity (higher pK _a)	Alkyl groups push electrons toward O, strengthening O-H bond — harder to release H ⁺	(CH ₃) ₃ COH least acidic; CH ₃ OH most acidic
EWG near -OH	INCREASE acidity (lower pK _a)	EWG pulls electron density away from O, destabilising alkoxide less effectively — H releases more easily	CF ₃ CH ₂ OH (pK _a 12.4) vs CH ₃ CH ₂ OH (pK _a 16)
Phenol vs aliphatic alcohol	Phenol MUCH more acidic (pK _a 10)	Phenoxide anion is resonance-stabilised across benzene ring — much more stable than alkoxide	C ₆ H ₅ OH (pK _a 10) vs ROH (pK _a 16-18)

All Reactions — Conditions, Reagents, Products

Reaction	Reagent(s)	Conditions	Product(s)	Things to Note
Oxidation of 1 degree to aldehyde	PCC (pyridinium chlorochromate) or controlled $K_2Cr_2O_7/H_2SO_4$	Mild: PCC in CH_2Cl_2 at RT. $K_2Cr_2O_7$: distil off aldehyde immediately	Aldehyde ($RCHO$)	PCC stops at aldehyde stage. $K_2Cr_2O_7$ with excess goes all the way to acid.
Oxidation of 1 degree to carboxylic acid	Excess $K_2Cr_2O_7/H_2SO_4$ or $KMnO_4/H^+$	Reflux with excess oxidant	Carboxylic acid ($RCOOH$)	$K_2Cr_2O_7$ changes orange to green. $KMnO_4$ changes purple to colourless/brown. Both are visual colour changes.
Oxidation of 2 degree	$K_2Cr_2O_7/H_2SO_4$ or $KMnO_4$	Heat/reflux	Ketone ($RCOR$) — stops here	No further oxidation — ketone has no H on the carbonyl carbon.
Oxidation of 3 degree	$K_2Cr_2O_7$ or $KMnO_4$	Any conditions	No reaction	No H on the C bearing -OH — cannot be oxidised without breaking a C-C bond.
Dehydration to alkene	Conc. H_2SO_4	170 degrees C — HIGH temperature	Alkene + H_2O	Zaitsev rule: most substituted alkene is major product. E2 mechanism.
Dehydration to ether	Conc. H_2SO_4	140 degrees C — LOWER temperature	Ether (ROR) + H_2O	Two alcohol molecules combine. Only works well for 1 degree alcohols.
Fischer esterification	$RCOOH$ + cat. H_2SO_4	Heat, reflux — reversible equilibrium	Ester ($RCOOR$) + H_2O	Push right: excess alcohol or remove water. H_2SO_4 is catalyst only — not consumed.

Reaction	Reagent(s)	Conditions	Product(s)	Things to Note
Lucas test (distinguish types)	ZnCl ₂ + Conc. HCl (Lucas reagent)	Room temperature — NO heating	Alkyl chloride + H ₂ O (turbidity = cloudiness)	3 degree = IMMEDIATE turbidity. 2 degree = turbidity in 5 min. 1 degree = NO reaction at RT. Only for alcohols up to 6 carbons.
With Na metal	Sodium metal Na	Room temperature — no water present	Alkoxide salt (RONa) + H ₂ gas (fizzing)	All alcohols react but SLOWER than Na + water. Fizzing confirms H ₂ gas. 3 degree reacts slowest.
With SOCl ₂ (cleanest route to RCl)	Thionyl chloride SOCl ₂	Pyridine solvent, room temperature	Alkyl chloride (RCl) + SO ₂ gas + HCl gas	Cleanest: byproducts SO ₂ and HCl are gases — easily removed. No rearrangement. Pyridine absorbs HCl.
With PCl ₅	Phosphorus pentachloride PCl ₅	Room temperature	Alkyl chloride (RCl) + POCl ₃ + HCl	Less clean than SOCl ₂ — POCl ₃ is a liquid byproduct and harder to remove.
With PBr ₃	Phosphorus tribromide PBr ₃	Room temp or mild heat	Alkyl bromide (RBr) + H ₃ PO ₃	S _N 2 mechanism — INVERSION of configuration at chiral center. Used for making RBr.

THINGS TO NOTE: THINGS TO NOTE: (1) Phenols react with NaOH (unlike aliphatic alcohols which need Na metal). (2) Phenols react with Na₂CO₃ but NOT NaHCO₃. (3) Tertiary alcohols cannot be oxidised under normal conditions — no H on the C-OH. (4) Lucas test only reliable for alcohols with 6 or fewer carbons. (5) Order of reactivity with HX: 3 degree > 2 degree > 1 degree (mirrors carbocation stability). (6) Methanol is TOXIC — metabolised to formaldehyde causing blindness.

EXAM HACK: Oxidation: 1 degree to aldehyde (mild/PCC) to acid (excess K₂Cr₂O₇). 2 degree to ketone (stops). 3 degree = NOTHING. Lucas: 3 degree = instant cloud. 2 degree = 5 min. 1 degree = no reaction at RT. SOCl₂ = cleanest route to alkyl chloride.

ALZHEIMER'S MODE: Oxidation is like burning a candle. 1 degree burns all the way to an acid. 2 degree burns halfway to a ketone and stops. 3 degree won't even light. Lucas test: cloudiness =

alcohol converting to insoluble alkyl chloride. More branches = faster reaction.

CHAPTER 8 — ETHERS

Ethers have formula R-O-R'. Generally CHEMICALLY INERT and widely used as solvents.

Property or Reaction	Result	Reagents and Conditions	Things to Note
Boiling point	Much LOWER than alcohols of same MW	—	No O-H H-bonding between ether molecules. Similar BP to alkanes of same MW.
Solubility in water	Slightly soluble (low MW ethers)	—	O lone pairs can accept H-bonds FROM water. Cannot donate H-bonds.
Flammability	Highly FLAMMABLE — fire risk	—	Very low BP; vapour denser than air and spreads along the floor. Flash point of diethyl ether = -45 degrees C.
Peroxide formation — DANGER	Explosive organic peroxides form on standing	Air/O ₂ — spontaneous in light or heat	ALWAYS test old ether bottles with peroxide test strip before heating. Peroxides can detonate on distillation!
Cleavage (1 equiv strong acid)	Alkyl halide + alcohol	Conc. HI or conc. HBr; heat required	Order: HI > HBr >> HCl. S _N 2 for 1 degree/2 degree ethers; S _N 1 for 3 degree ethers. HCl alone is too weak.
Cleavage (excess HI or HBr)	Two molecules of alkyl halide	Excess HI or HBr; heat	Both C-O bonds cleaved. Alcohol intermediate reacts with second HX molecule.
Reaction with conc. H ₂ SO ₄	Protonation of O — dissolves	Conc. H ₂ SO ₄ only	Used to separate ethers from alkanes. Ether dissolves in H ₂ SO ₄ ; alkane does not.
No reaction with:	No product — ether is inert	NaOH, KOH, KMnO ₄ , LiAlH ₄ , NaBH ₄ , K ₂ Cr ₂ O ₇	This inertness makes ethers IDEAL solvents for Grignard reactions and LiAlH ₄ reductions.

THINGS TO NOTE: THINGS TO NOTE: (1) IUPAC name: smaller alkyl group = alkoxy prefix; longer chain = parent chain. E.g. CH₃-O-CH₂CH₃ = methoxyethane. (2) Common name: name both alkyl groups alphabetically then add 'ether'. E.g. methyl ethyl ether. (3) Grignard reactions MUST be run in ether solvents (THF or diethyl ether) — the O lone pairs stabilise the Mg centre. (4) Cyclic ethers (THF, dioxane) are more water-soluble than acyclic ethers. (5)

Epoxides (3-membered cyclic ethers) are VERY reactive unlike acyclic ethers — see next chapter.

EXAM HACK: Ethers resist NaOH, KMnO₄, LiAlH₄, NaBH₄. ONLY strong acids (HI > HBr) cleave them. Inert = perfect solvent for sensitive reactions. **PEROXIDE DANGER:** always check old ether bottles!

ALZHEIMER'S MODE: Ether = lazy, stubborn molecule that refuses to react with almost anything. Only the strongest acids (HI) can force it open. Perfect solvent because it minds its own business and does not interfere.

CHAPTER 9 — EPOXIDES (OXIRANES)

Epoxides = cyclic ethers with a 3-membered ring containing oxygen. Ring strain (~115 kJ/mol) makes them HIGHLY REACTIVE.

Preparation

Method	Reagent	Conditions	Product	Notes
Oxidation of alkene with peracid	m-CPBA or RCOOOH (peracid)	RT, CH ₂ Cl ₂ solvent	Epoxide (3-membered ring)	Syn addition: O delivered from same face — stereochemistry of alkene RETAINED in epoxide.
From halohydrin + base	Step 1: X ₂ /H ₂ O gives halohydrin. Step 2: NaOH (base)	Step 1: RT. Step 2: dilute base	Epoxide via intramolecular SN2	-OH acts as internal nucleophile displacing halide. Overall anti addition.

Ring-Opening Reactions — Full Guide

Condition	Reagent	Attack at...	Mechanism	Product	Stereochemistry	Key Note
Acid-catalyzed	H ₃ O ⁺ + Nu (H ₂ O, ROH, HX)	MORE substituted carbon	SN1-like: partial carbocation character at more substituted C	Trans-1,2-diol, trans-halohydrin, etc.	ANTI addition (trans)	H ⁺ protonates epoxide O making it better electrophile. More substituted = more stable partial carbocation. Nu attacks from OPPOSITE side.
Base-catalyzed	-OH, -CN, RMgX, LiAlH ₄ , RNH ₂	LESS substituted carbon	SN2: backside attack at less hindered C	Trans-diol, trans-cyanohydrin, alcohol, amino alcohol	ANTI addition (trans)	No carbocation. Nu attacks where there is LESS steric hindrance. Still anti because Nu enters from back.
With H ₂ O + H ⁺	H ₂ SO ₄ /H ₂ O	More substituted (acid)	SN1-like	1,2-Diol (glycol)	Trans (anti)	Industrial: ethylene oxide + H ₂ O/H ⁺ = ethylene glycol (antifreeze for cars).
With LiAlH ₄	H ⁻ hydride source	LESS substituted	SN2	Alcohol at less substituted C	Inversion at that C	Strong reducing agent. H ⁻ is strong Nu. Goes to less hindered end.

Condition	Reagent	Attack at...	Mechanism	Product	Stereochemistry	Key Note
With Grignard RMgX	R ⁻ carbanion	LESS substituted	SN2	Alcohol with new C-C bond at less substituted end	Inversion at that C	Excellent for making NEW C-C BONDS. Extends carbon chain.

THINGS TO NOTE: THINGS TO NOTE: (1) BOTH acid and base catalyzed opening give ANTI (trans) products. Nu always attacks from OPPOSITE side to oxygen. (2) What CHANGES between acid and base: the REGIOCHEMISTRY. Acid = more substituted end. Base = less substituted end. (3) Ethylene oxide (simplest epoxide) + H₂O/H⁺ = ethylene glycol (antifreeze). (4) Grignard + epoxide = excellent way to add a new carbon chain with an alcohol at the end.

EXAM HACK: Acid = attack MORE substituted. Base = attack LESS substituted. BOTH give anti (trans) product. Grignard + epoxide = new C-C bond at the LESS substituted end.

ALZHEIMER'S MODE: Epoxide = tiny stressed ring ready to snap open. Acid pulls it open at the fat end (more substituted). Base opens it at the thin end (less substituted). Nucleophile always enters from opposite side = trans/anti product every time.

CHAPTER 10 — CARBOXYLIC ACIDS AND DERIVATIVES

Carboxylic acids contain the -COOH group. pK_a approx 4 to 5. The carboxylate anion RCOO^- is stabilised by resonance across BOTH oxygens — this is why they are much stronger acids than alcohols.

Acidity Comparison

Compound	pK_a	Strength	Key Reason
HCl	-7	Mineral acid	Complete ionisation
CF_3COOH (trifluoroacetic)	0	Strongest org. acid	3x F atoms = very strong EWG — maximum inductive withdrawal
CCl_3COOH (trichloroacetic)	0.66	Very strong	3x Cl = strong EWG inductive effect
CHCl_2COOH (dichloroacetic)	1.48	Strong	2x Cl
CH_2ClCOOH (chloroacetic)	2.86	Moderate-strong	1x Cl
HCOOH (formic acid)	3.75	Stronger than acetic	No alkyl EDG to destabilise carboxylate
$\text{C}_6\text{H}_5\text{COOH}$ (benzoic)	4.20	Similar to acetic	Benzene ring mildly EWG overall
CH_3COOH (acetic) — REFERENCE	4.75	Moderate	1 methyl group EDG slightly destabilises
$\text{C}_6\text{H}_5\text{OH}$ (phenol)	10.0	Much weaker	Phenoxide resonance-stabilised but less so
ROH (alcohols)	16 to 18	Very weak	No resonance stabilisation of alkoxide
RH (alkanes)	~50	Extremely weak	Carbanion is very unstable

Reactions of Carboxylic Acids

Reaction	Reagent(s)	Conditions	Product(s)	Things to Note
Salt formation	NaOH, Na ₂ CO ₃ , or NaHCO ₃	Aqueous, room temperature	Carboxylate salt + H ₂ O; CO ₂ gas with Na ₂ CO ₃ /NaHCO ₃	NaHCO ₃ + RCOOH = effervescence (CO ₂). Phenols do NOT react with NaHCO ₃ — use this to distinguish carboxylic acids from phenols!
Esterification (Fischer)	R'OH + cat. H ₂ SO ₄ or HCl	Heat, reflux — reversible equilibrium	Ester (RCOOR') + H ₂ O	Push right: excess alcohol or remove water as it forms. H ₂ SO ₄ is catalyst only.
Acid chloride (RCOCl)	SOCl ₂ (preferred) or PCl ₅	Gentle heat, anhydrous	RCOCl + SO ₂ + HCl (with SOCl ₂)	SOCl ₂ preferred: SO ₂ and HCl are gases — easily removed. Most reactive acid derivative. Fumes in moist air.
Anhydride	Two RCOOH molecules or P ₂ O ₅	Heat — dehydration	(RCO) ₂ O + H ₂ O	Mixed anhydrides from two different acids (use acid chloride + carboxylate salt).
Reduction to 1 degree alcohol	LiAlH ₄	Dry ether, 0 degrees C then reflux	Primary alcohol (RCH ₂ OH)	NaBH ₄ does NOT reduce COOH. ONLY LiAlH ₄ reduces carboxylic acids.
Decarboxylation	Heat	Strong heat above 150 degrees C; beta-keto acids decarboxylate easily via 6-membered TS	Alkane or ketone + CO ₂	Malonic acid: HOOCCH ₂ COOH gives CH ₃ COOH + CO ₂ easily.
Amide formation	NH ₃ then heat	Forms ammonium salt first; heat dehydrates to amide	Amide (RCONH ₂) + H ₂ O	Heat is essential to drive off water and convert ammonium salt to amide.

Reactivity of Acid Derivatives

Rank	Derivative	Leaving Group	Why	Hydrolysis Rate
1 MOST reactive	Acid chloride (RCOCl)	Cl ⁻ (excellent LG)	Cl ⁻ is one of the best leaving groups; C=O is very electrophilic	Fumes in moist air; reacts with water instantly
2	Anhydride (RCOOCOR)	RCOO ⁻ (decent LG)	Carboxylate is a moderate leaving group	Slow reaction with water on standing
3	Ester (RCOOR')	RO ⁻ (poor LG)	Alkoxide is a poor leaving group	Needs acid/base catalyst + heat
4 LEAST reactive	Amide (RCONH ₂)	NH ₂ ⁻ / NH ₃ (terrible LG)	N lone pair donates into C=O (resonance) making C less electrophilic; NH ₂ ⁻ is a terrible leaving group	Vigorous conditions: prolonged reflux with conc. acid or base

THINGS TO NOTE: THINGS TO NOTE: (1) All acid derivative reactions proceed by **NUCLEOPHILIC ACYL SUBSTITUTION** — tetrahedral intermediate forms, then leaving group departs. (2) NaHCO₃ gives CO₂ with RCOOH but NOT with phenols, alcohols, or ketones — definitive test for carboxylic acids. (3) Dicarboxylic acids names: oxalic (2C), malonic (3C), succinic (4C), glutaric (5C), adipic (6C). (4) More EWG near -COOH = stronger acid (lower pKa). More EDG = weaker acid.

EXAM HACK: EWG near -COOH = stronger acid (lower pKa). NaHCO₃ gives CO₂ with RCOOH only — not phenols. Reactivity: Acid chloride > Anhydride > Ester > Amide. LiAlH₄ reduces all. NaBH₄ does NOT reduce COOH.

ALZHEIMER'S MODE: Carboxylic acid is strong because after releasing H⁺, the negative charge spreads over TWO oxygens (resonance) — like two people sharing a heavy bag. More EWG nearby = more help pulling electrons away = easier to release H⁺ = stronger acid.

CHAPTER 11 — ESTERS

Esters have formula RCOOR' . Pleasant fruity smells. Used in flavourings, perfumes, biodiesel, and polymers.

Formation Methods

Method	Reagents	Conditions	Product	Things to Note
Fischer Esterification	$\text{RCOOH} + \text{R}'\text{OH}$	Cat. H_2SO_4 or HCl ; heat and reflux	$\text{RCOOR}' + \text{H}_2\text{O}$	REVERSIBLE. Push right: excess alcohol or remove water with Dean-Stark trap.
From acid chloride + alcohol	$\text{RCOCl} + \text{R}'\text{OH}$	Base (pyridine) to absorb HCl ; RT	$\text{RCOOR}' + \text{HCl}$	FAST and IRREVERSIBLE. Pyridine drives reaction by absorbing HCl .
From anhydride + alcohol	$(\text{RCO})_2\text{O} + \text{R}'\text{OH}$	Mild conditions; sometimes acid cat.	$\text{RCOOR}' + \text{RCOOH}$	Used for acetylation. Aspirin made from acetic anhydride + salicylic acid.
Transesterification	$\text{RCOOR}' + \text{R}''\text{OH}$	Acid or base catalyst; heat	$\text{RCOOR}'' + \text{R}'\text{OH}$	Exchange of alkoxy group. How biodiesel is made: vegetable oil + methanol gives FAME.

Hydrolysis of Esters

Type	Conditions	Products	Reversible?	Key Note
Acid hydrolysis	$\text{H}_2\text{O} + \text{H}^+$ (dil. H_2SO_4); heat	Carboxylic acid + Alcohol	YES (reversible)	Reverse of Fischer. Use excess water to push right.
Base hydrolysis (Saponification)	$\text{NaOH} + \text{H}_2\text{O}$; heat	Carboxylate SALT + Alcohol	NO (irreversible)	Irreversible — salt will not react back with alcohol. Used to make SOAP from fats.
Aminolysis	NH_3 or RNH_2	Amide + Alcohol	Slow, hard to reverse	Useful route to amides without strong acid conditions.
Reduction with LiAlH_4	LiAlH_4 ; dry ether	TWO primary alcohols (one from each half of ester)	N/A	NaBH_4 does NOT reduce esters. LiAlH_4 gives 2 alcohols.

Common Fruity Esters

Ester	Smell
Ethyl ethanoate (ethyl acetate)	Pear and nail polish remover
Isoamyl acetate (3-methylbutyl ethanoate)	Banana
Methyl butanoate	Pineapple and apple
Octyl acetate	Orange
Methyl salicylate	Wintergreen and deep heat
Benzyl acetate	Jasmine

THINGS TO NOTE: THINGS TO NOTE: (1) Saponification = NaOH hydrolysis of fats (triglycerides) = soap making. Fat + 3 NaOH gives 3 soap molecules + glycerol. (2) Soap = sodium carboxylate salt: ionic head (water-loving) + long carbon tail (oil-loving). Forms micelles to trap grease. (3) Biodiesel = FAME (fatty acid methyl esters) from transesterification of vegetable oils with methanol. (4) Aspirin = acetylsalicylic acid — an ester hydrolysed in the body to salicylic acid and acetic acid. (5) Polyester (e.g. Terylene/Dacron) = condensation polymer of a diacid + diol.

EXAM HACK: Ester + NaOH = saponification (IRREVERSIBLE, gives salt). Ester + H₂O/H⁺ = reversible hydrolysis (gives acid). LiAlH₄ reduces ester to 2 alcohols. NaBH₄ does NOT reduce esters.

ALZHEIMER'S MODE: Esters smell like fruits. Acid + alcohol combine and drop water = ester. NaOH breaks it permanently (soap). H₂SO₄ + water breaks it but it reforms. Soap: one end loves water, one end loves oil — cleans grease from hands.

CHAPTER 12 — ALDEHYDES AND KETONES

Both contain the carbonyl group $C=O$. ALDEHYDE: $-CHO$ at chain end. KETONE: $C=O$ in chain middle. Key difference: aldehyde H can be oxidised; ketone cannot easily be oxidised.

Distinguishing Tests — Complete Reference

Test	Reagent	Aldehyde Result	Ketone Result	Key Notes and Conditions
Tollens (silver mirror)	$AgNO_3$ in excess NH_3 — Tollens reagent	POSITIVE: Silver mirror on glass	NEGATIVE: No reaction	Works for ALL aldehydes including benzaldehyde. Warm gently — NEVER boil (explosion risk from silver nitride). Test tube must be clean and grease-free.
Fehling's	Fehling's A ($CuSO_4$ aq) + Fehling's B ($NaOH$ + sodium potassium tartrate) — heat after mixing	POSITIVE: Brick-red precipitate (Cu_2O)	NEGATIVE: Stays blue	ONLY aliphatic aldehydes. Benzaldehyde = NEGATIVE. Reducing sugars also positive. Cu^{2+} reduced to Cu^+ (Cu_2O brick-red solid).
Benedict's	Benedict's reagent (blue: $CuSO_4$ + Na_2CO_3 + sodium citrate)	POSITIVE: Orange-red precipitate	NEGATIVE: Stays blue	More stable than Fehling's (one bottle). Used clinically to detect glucose in urine (glycosuria test).
2,4-DNPH (Brady's reagent)	2,4-Dinitrophenylhydrazine (orange solution)	POSITIVE: Orange or yellow precipitate	POSITIVE: Orange or yellow precipitate	BOTH aldehydes AND ketones test positive — detects the $C=O$ group. The melting point of the precipitate identifies the specific compound.
Iodoform test	I_2 + $NaOH$ (aqueous)	POSITIVE only for CH_3CHO (ethanal)	POSITIVE only for methyl ketones CH_3COR	Yellow precipitate (CHI_3) with distinctive smell. Also positive for ethanol and secondary alcohols with CH_3CHOH- group (these are first oxidised to CH_3CO- group).
Schiff's test	Fuchsin dye decolorised with SO_2	POSITIVE: Pink or magenta colour returns	NEGATIVE: Stays colourless	More sensitive than Tollens. Formaldehyde gives permanent colour even without warming.

Test	Reagent	Aldehyde Result	Ketone Result	Key Notes and Conditions
KMnO ₄ (Baeyer's test)	Dilute purple KMnO ₄	POSITIVE: Purple decolorised to brown	NEGATIVE: Purple colour stays	Aldehydes reduce KMnO ₄ . Not specific — alkenes and alkynes also decolorise it. Use with other tests.

All Reactions — Conditions, Reagents, Products

Reaction	Reagent(s)	Conditions	Product	Aldehyde/Ketone ?	Things to Note
Nucleophilic addition of HCN	HCN or KCN + H ₂ SO ₄	RT; slightly acidic or basic	Cyanohydrin (alpha-hydroxy nitrile)	Both	Adds one carbon. Cyanohydrins hydrolysed to alpha-hydroxy acids (useful!).
Reduction to alcohol	NaBH ₄ (mild) or LiAlH ₄ (strong)	RT for NaBH ₄ ; dry ether 0 degrees for LiAlH ₄	Aldehyde gives 1 degree alcohol; Ketone gives 2 degree alcohol	Both	NaBH ₄ selectively reduces C=O but NOT COOH, COOR. LiAlH ₄ reduces all carbonyls.
Oxidation	KMnO ₄ /H ⁺ or K ₂ Cr ₂ O ₇ /H ⁺ or Tollens or Fehling	Heat for K-permanganate/dichromate; gentle warm for Tollens/Fehling	Carboxylic acid (RCOOH)	ALDEHYDES ONLY (easily oxidised)	Ketones resist mild oxidation. Strong KMnO ₄ cleaves the C-C bonds of ketones.
Acetal formation	Excess ROH + cat. H ⁺	Dry HCl or H ₂ SO ₄ ; anhydrous conditions essential	Hemiacetal then Acetal + H ₂ O	Aldehydes mainly (ketones give ketals)	Acetal is base-stable. Hydrolysed by dilute acid. Used to PROTECT aldehyde group during synthesis.
Aldol condensation	Dilute NaOH (base)	RT gives aldol product; heat gives alpha,beta-unsaturated C=O	Beta-hydroxy aldehyde/ketone then dehydration product with heat	Both (need at least one alpha-H)	Base removes alpha-H to form enolate; enolate attacks another C=O. Requires alpha-H.

Reaction	Reagent(s)	Conditions	Product	Aldehyde/Ketone ?	Things to Note
Wittig reaction	Phosphorus ylide $\text{Ph}_3\text{P}=\text{CR}_2$	RT; organic solvent	Alkene ($\text{C}=\text{O}$ replaced by $\text{C}=\text{C}$) + $\text{Ph}_3\text{P}=\text{O}$	Both	Converts $\text{C}=\text{O}$ SPECIFICALLY to $\text{C}=\text{C}$. No ambiguity. Useful in targeted synthesis.
Cannizzaro reaction	Conc. NaOH	Concentrated NaOH; heat	One aldehyde oxidised to carboxylate + one reduced to alcohol	Aldehydes with NO α -H only (HCHO, PhCHO)	Disproportionation: one molecule oxidised, one reduced simultaneously. No external oxidant/reductant.
Clemmensen reduction	Zn-Hg amalgam + conc. HCl	Heat; strongly acidic	$\text{C}=\text{O}$ reduced to CH_2 (removes oxygen completely)	Both	For acid-stable substrates. Converts ketone directly to methylene (alkane-like C).
Wolff-Kishner reduction	NH_2NH_2 hydrazine then KOH	Heat; basic conditions; ethylene glycol solvent	$\text{C}=\text{O}$ reduced to CH_2	Both	For base-stable substrates. Alternative to Clemmensen.

THINGS TO NOTE: THINGS TO NOTE: (1) Fehling's = aliphatic aldehydes ONLY (benzaldehyde NEGATIVE). Tollens = ALL aldehydes (including benzaldehyde POSITIVE). (2) Iodoform test is positive for: ethanal, methyl ketones (CH_3COR), ethanol, $\text{CH}_3\text{CHOH-}$ group. (3) 2,4-DNP precipitate melting point is used in TABLES to identify the specific aldehyde or ketone. (4) Alpha-hydrogen (adjacent to $\text{C}=\text{O}$) has $\text{pK}_a \sim 20$ — more acidic than ordinary C-H due to enolate resonance stabilisation. (5) Aldol condensation requires at least ONE reactant to have an α -H.

EXAM HACK: Tollens = silver mirror = ALDEHYDE ONLY. DNPH = orange ppt = BOTH aldehyde AND ketone. Fehling = brick red = aliphatic aldehyde ONLY. Iodoform = yellow CHI_3 = methyl ketone or ethanal. NaBH_4 reduces $\text{C}=\text{O}$ only. LiAlH_4 reduces everything.

ALZHEIMER'S MODE: Aldehyde = pipe open at the end (CHO) — gets easily oxidised to acid. Ketone = sealed in the middle — hard to oxidise. Tollens silver mirror = aldehyde confirmed. NaBH_4 is gentle. LiAlH_4 is the aggressive one that reduces everything.

CHAPTER 13 — AMINES

Amines = organic derivatives of NH_3 where H atoms are replaced by organic groups. BASIC due to lone pair on N.

Classification and Basicity

Class	Structure	pK _b (approx)	Base Strength	Reason
Aliphatic 2 degree (R ₂ NH)	R ₂ NH	~3.3	Strongest aliphatic	2 alkyl groups donate electrons; good solvation of R ₂ NH ₂ ⁺ in water
Aliphatic 1 degree (RNH ₂)	RNH ₂	~3.4	Very strong	1 alkyl group donates electrons to N
Aliphatic 3 degree (R ₃ N)	R ₃ N	~4.2	Strong but less than 2 degree	3 alkyl groups donate but steric hindrance reduces solvation of R ₃ NH ⁺
Ammonia NH ₃	NH ₃	4.74	Moderate — reference	No alkyl groups to donate electrons
Aromatic amines (ArNH ₂ , aniline)	ArNH ₂	~9.4	Weak	Lone pair on N delocalised INTO benzene ring = less available to accept H ⁺
Amides (RCONH ₂)	RCONH ₂	~15	Extremely weak (barely basic)	Lone pair on N delocalised to C=O group = almost completely unavailable for accepting H ⁺
ArNH ₂ with EWG on ring (e.g. -NO ₂)	p-NO ₂ -C ₆ H ₄ -NH ₂	Higher pK _b	Weaker than aniline	EWG further reduces electron density available on N
ArNH ₂ with EDG on ring (e.g. -CH ₃)	p-CH ₃ -C ₆ H ₄ -NH ₂	Lower pK _b	Stronger than aniline	EDG feeds more electron density towards ring and N

All Reactions of Amines

Reaction	Reagent(s)	Conditions	Product	Class	Things to Note
Salt formation	HCl, HBr, HNO ₃ , H ₂ SO ₄	Aqueous; room temp	Ammonium salt (RNH ₃ ⁺ X ⁻)	All (1, 2, 3 degree)	Ionic salts are water-soluble. Convert back to amine with NaOH.

Reaction	Reagent(s)	Conditions	Product	Class	Things to Note
Alkylation	Alkyl halide RX	Heat; SN2 conditions	Higher amine (1 to 2 to 3 to quaternary salt)	1, 2, 3 degree	Gives MIXTURE of products (over-alkylation). Use Gabriel synthesis for pure 1 degree amine.
Acylation	Acid chloride RCOCl or anhydride (RCO) ₂ O	Base (pyridine) to absorb HCl; RT to mild heat	Amide (RCONHR') + HCl or + RCOOH	1 and 2 degree only (3 degree has no N-H)	Used to PROTECT amino groups in synthesis. Amide is less nucleophilic and less basic.
Diazotization (1 degree aromatic only)	NaNO ₂ + HCl (gives HNO ₂ in situ)	MUST STAY AT 0 to 5 degrees C — critical!	Diazonium salt ArN ₂ ⁺ Cl ⁻	Primary AROMATIC amines ONLY	Above 5 degrees C = diazonium decomposes to phenol. Aliphatic 1 degree amines form unstable diazonium that decomposes immediately.
Gabriel synthesis	Phthalimide + KOH; then RX; then hydrazine or NaOH hydrolysis	Reflux each step	PURE primary amine (RNH ₂) with NO 2 degree or 3 degree contamination	Gives PURE 1 degree amine only	Clean synthesis. Phthalimide acts as the N-H source that then gets alkylated and released.
Hofmann degradation	Amide RCONH ₂ + Br ₂ + NaOH	Heat	Primary amine with ONE LESS CARBON than the amide + CO ₂	Primary amides only	Carbon count DECREASES by 1. Mechanism: N-bromo amide forms, rearranges to isocyanate, then hydrolysis.
Carbylamine test	CHCl ₃ + alcoholic KOH	Heat gently	Isocyanide (R-N triple bond C) — unbearable foul smell	PRIMARY amines ONLY	VERY specific. 2 degree and 3 degree = NO reaction. Definitive test for 1 degree amines in exams.
With HNO ₂ : 1 degree aliphatic	NaNO ₂ + HCl	0 to 5 degrees C	Unstable diazonium gives N ₂ gas + alcohol/alkene mixture	Aliphatic 1 degree	Not useful synthetically. N ₂ gas evolution confirms 1 degree aliphatic amine.

Reaction	Reagent(s)	Conditions	Product	Class	Things to Note
With HNO ₂ : 2 degree amine	NaNO ₂ + HCl	0 to 5 degrees C	N-Nitrosoamine (R ₂ N-N=O) — YELLOW oily product	Secondary amines	N-nitrosamines are carcinogens. Yellow oil is characteristic of 2 degree amines with HNO ₂ .
With HNO ₂ : 3 degree amine	NaNO ₂ + HCl	0 degrees C	Ammonium salt only — no N-nitroso product	Tertiary amines	No N-H available for reaction.

Diazonium Salt Replacement Reactions

Reagent	Product	Reaction Name	Conditions	Notes
H ₂ O then heat	ArOH (phenol)	Hydrolysis	Warm dilute acid	H ₂ O displaces N ₂
HBF ₄ then dry heat	ArF (aryl fluoride)	Balz-Schiemann	Heat dry ArN ₂ +BF ₄ -salt	ONLY route to aryl fluorides
CuCl in HCl	ArCl (aryl chloride)	Sandmeyer	Warm	Cu(I) catalyst essential
CuBr in HBr	ArBr (aryl bromide)	Sandmeyer	Warm	Cu(I) catalyst essential
KI solution	ArI (aryl iodide)	—	Warm	No Cu needed — I- is good Nu
CuCN in KCN	ArCN (nitrile)	Sandmeyer	Warm	Extends C chain by 1 — very useful!
H ₃ PO ₂ /H ₂ O	ArH (deamination)	Deamination	Room temperature	Removes -N ₂ ⁺ and replaces with H
ArOH or ArNH ₂ in base	Ar-N=N-Ar' (azo dye)	Azo coupling	Cold 0 to 5 degrees C	Intensely coloured azo compounds; used in dyes

THINGS TO NOTE: THINGS TO NOTE: (1) Diazotization MUST be at 0-5 degrees C — above this, ArN₂⁺ decomposes to phenol. (2) Gabriel synthesis = PURE 1 degree amines only. This is its greatest advantage over direct alkylation. (3) Hofmann degradation LOSES ONE CARBON — amide has n+1 carbons, product amine has n carbons. (4) Carbylamine test = 1 degree only. 2 degree gives yellow oily N-nitroso compound with HNO₂ instead. (5) Aniline (aromatic amine) is easily oxidised in air — turns brown on standing.

EXAM HACK: CARBYLAMINE = 1 degree amines ONLY (foul isocyanide smell). GABRIEL = only 1 degree (pure product). HOFMANN = loses 1 carbon. DIAZOTIZATION = must stay COLD (0-5 C). Aniline = WEAK base (benzene steals the lone pair).

ALZHEIMER'S MODE: Amines = ammonia with carbon arms. They grab H^+ because N has a lone pair. Aromatic amines (aniline) are weak — benzene ring grabs the lone pair away from N. Carbylamine: if it smells horrible = primary amine confirmed. Gabriel = clean way to make primary amines without mess.

CHAPTER 14 — CARBOHYDRATES

Carbohydrates = polyhydroxy aldehydes or ketones. Empirical formula $C_n(H_2O)_n$. Most abundant organic compounds on Earth.

Classification and Glucose

Class	Size	Example	Function
Monosaccharide	1 unit — not hydrolysable	Glucose, Fructose, Galactose, Ribose	Energy; building blocks
Disaccharide	2 units — hydrolysable	Sucrose, Maltose, Lactose	Energy transport
Oligosaccharide	3 to 10 units	Raffinose, Stachyose	Cell signalling
Polysaccharide	100s to 1000s of units	Starch, Cellulose, Glycogen	Storage and structure

Glucose Fact	Detail
Formula	$C_6H_{12}O_6$ — aldohexose (6-carbon aldehyde sugar)
Chiral centers	4 (at C2, C3, C4, C5) — max $2^4 = 16$ stereoisomers theoretically
Ring closure	C-1 aldehyde reacts with C-5 -OH to form 6-membered glucopyranose ring
Anomeric carbon	C-1: new chiral center created during ring formation
alpha-glucose	OH at C-1 is AXIAL (pointing down in chair form)
beta-glucose	OH at C-1 is EQUATORIAL (pointing up in chair form — more stable)
At equilibrium	64% beta, 36% alpha — beta more stable due to equatorial OH
Mutarotation	alpha and beta forms interconvert via the open-chain form in solution

Key Disaccharides

Disaccharide	Units	Linkage	Reducing?	Enzyme	Key Notes
Sucrose (table sugar)	Glucose + Fructose	alpha(1 to 2)beta — head-to-head	NO	Sucrase	NON-reducing: BOTH anomeric carbons involved in glycosidic bond. Hydrolysis = invert sugar (glucose + fructose).
Maltose (malt sugar)	Glucose + Glucose	alpha(1 to 4)	YES	Maltase	Product of starch digestion by amylase. Free anomeric C at reducing end.
Lactose (milk sugar)	Galactose + Glucose	beta(1 to 4)	YES	Lactase	Lactase deficiency = lactose intolerance. Fermented to lactic acid (yoghurt/cheese).
Cellobiose	Glucose + Glucose	beta(1 to 4)	YES	Cellobiase	Repeating unit of cellulose. Humans cannot hydrolyse.
Trehalose	Glucose + Glucose	alpha(1 to 1)alpha — head-to-head	NO	Trehalase	Both anomeric C involved = non-reducing. Found in insects and fungi.

Colour Tests for Carbohydrates

Test	Reagent	Result	Detects	Important Notes
Molisch's	Conc. H ₂ SO ₄ + alpha-naphthol	Purple or violet ring at interface	ALL carbohydrates (general test)	Negative = compound is NOT a carbohydrate. Works for mono, di, and polysaccharides.
Fehling's	Fehling's A + B — blue solution	Brick-red precipitate (Cu ₂ O)	Reducing sugars (free anomeric -OH)	Negative for sucrose and trehalose (non-reducing). Positive for glucose, maltose, lactose, fructose.
Benedict's	Benedict's reagent (blue)	Orange-red precipitate	Reducing sugars	Used clinically for glucose in urine. More convenient than Fehling's (one bottle).
Tollens'	AgNO ₃ in excess NH ₃	Silver mirror on glass	Reducing sugars	Fructose (ketose) also tests positive — enolises in alkaline conditions.

Test	Reagent	Result	Detects	Important Notes
Seliwanoff's	Resorcinol + conc. HCl; heat	Cherry-red colour FAST	KETOSES (fructose) specifically	Fructose = cherry-red in 2 min. Aldoses (glucose) = only faint pink after 10+ min. Distinguishes ketoses from aldoses.
Bial's	Orcinol + conc. HCl + FeCl ₃	Blue-green colour	PENTOSES (5-carbon sugars)	Ribose = positive. Hexoses give brownish colour. Detects RNA component ribose.
Barfoed's	Copper acetate + acetic acid (mildly acidic)	Red precipitate within 5 min	Monosaccharides vs disaccharides	Monosaccharides reduce in 5 min. Disaccharides need much longer. Distinguishes mono from disaccharides.
Iodine / KI	I ₂ dissolved in KI solution	Blue-BLACK colour	Starch (amylose) SPECIFICALLY	Amylose helix traps I ₂ inside. Amylopectin = blue-purple. Glycogen = reddish-brown. Cellulose = no colour.

THINGS TO NOTE: THINGS TO NOTE: (1) REDUCING SUGAR = has free anomeric -OH (hemiacetal form) = can be oxidised. (2) NON-REDUCING = anomeric C involved in glycosidic bond (sucrose, trehalose). (3) Fructose is a KETOSE but is still a reducing sugar (enolises in alkaline conditions). (4) Invert sugar = sucrose hydrolysis product = 50:50 glucose + fructose — rotates polarised light in opposite direction to sucrose. (5) Mutarotation: alpha and beta interchange via the open chain in solution. (6) Specific rotation: alpha-glucose = +112 degrees; beta = +18.7; equilibrium mixture = +52.7 degrees.

EXAM HACK: Reducing sugar = free anomeric -OH = positive Tollens/Fehling/Benedict. Sucrose = NON-reducing (locked together). Seliwanoff's = cherry red = KETOSE. Bial's = blue-green = PENTOSE. Iodine = blue-black = STARCH (amylose) ONLY.

ALZHEIMER'S MODE: Glucose = the body's petrol. Ring forms: alpha or beta. alpha-chains = starch (you digest it). beta-chains = cellulose (you cannot). Sucrose = two sugars locked together through their active ends — cannot react with Fehling's. Mutarotation = glucose flipping between its ring forms in water.

CHAPTER 15 — POLYSACCHARIDES

Polysaccharides = large polymers of monosaccharides linked by glycosidic bonds. Energy storage (starch, glycogen) and structural support (cellulose, chitin).

Polysaccharide	Monomer	Linkage	Branching	Function	Iodine Test	Key Notes
Amylose (part of starch)	alpha-D-glucose	alpha(1 to 4) only	None — linear helix	Energy storage in plants	Deep BLUE-BLACK	~20-25% of starch. I ₂ molecules fit inside the helix.
Amylopectin (part of starch)	alpha-D-glucose	alpha(1 to 4) + alpha(1 to 6)	Yes — every 24-30 units	Energy storage in plants	Blue-purple	~75-80% of starch. Too branched for tight helix.
Glycogen	alpha-D-glucose	alpha(1 to 4) + alpha(1 to 6)	Highly branched every 8-12 units	Energy storage in animals (liver and muscle)	Reddish-BROWN (NOT blue)	More branched than amylopectin. Many chain ends allow rapid glucose release.
Cellulose	beta-D-glucose	beta(1 to 4) only	None — straight rigid chains	Structural: plant cell walls	No colour	Humans CANNOT digest. Parallel chains form H-bonds: extremely tough.
Chitin	N-acetyl-D-glucosamine	beta(1 to 4)	None	Exoskeleton of arthropods; fungal cell walls	No colour	2nd most abundant polysaccharide. Stronger than cellulose.

Alpha vs Beta Linkages

- alpha(1 to 4): chain coils into a helix = DIGESTIBLE by alpha-amylase = energy food for humans
- beta(1 to 4): chains lie flat in parallel = rigid and tough = INDIGESTIBLE by humans (dietary fibre)
- Starch (alpha) traps I₂ inside the helix = blue-black colour with iodine
- Cellulose (beta) has no helix = no I₂ trapping = no colour with iodine
- Glycogen is MORE branched than amylopectin = shorter helical sections = reddish-brown NOT blue with I₂

THINGS TO NOTE: THINGS TO NOTE: (1) Amylose = linear starch = deep blue-black with I₂. Amylopectin = branched starch = blue-purple. Glycogen = reddish-brown. Cellulose = colourless. (2) Salivary and pancreatic amylase hydrolyses alpha(1 to 4) bonds of starch to produce maltose. (3) Ruminants (cows, sheep) have cellulase-producing bacteria in their gut and CAN digest cellulose. Humans cannot. (4) Glycogen is stored in the liver and mobilised when blood glucose falls between meals. (5) Cellulose is used to make paper, cotton, and

rayon. Cellulose acetate = photographic film and spectacle frames.

EXAM HACK: alpha-linkage = digestible energy food. beta-linkage = indigestible structural fibre. Iodine: amylose = blue-black; amylopectin = blue-purple; glycogen = reddish-brown; cellulose = no colour. Starch = plant storage. Glycogen = animal storage (more branched).

ALZHEIMER'S MODE: alpha bonds = soft springs you can digest. beta bonds = stiff rods your body cannot break. Starch helix traps I₂ inside like a guest entering a spiral staircase = blue-black. Cellulose is flat planks stacked together = no room for I₂ = no colour.

CHAPTER 16 — PROTEINS

Proteins = polymers of L-alpha-amino acids joined by peptide bonds (-CO-NH-). They are the molecular machines of the cell.

Amino Acids and Structure Levels

Level	Name	Description	Bonds Responsible	Disrupted By
—	Amino acids	H ₂ N-CHR-COOH. Zwitterion at pH 7: +H ₃ N-CHR-COO ⁻ . pI = pH of zero net charge.	Covalent bonds in structure	Extreme pH (acid or base) fully hydrolyses
1 degree	Primary structure	Specific sequence of amino acids in the chain	Peptide bonds (covalent — very strong)	Acid/base hydrolysis — breaks peptide bonds
2 degree	Secondary structure	Local folding: alpha-helix or beta-pleated sheet	H-bonds between N-H and C=O of peptide bonds	Disruption of H-bonds by urea, heat, SDS
3 degree	Tertiary structure	Overall 3D shape of entire polypeptide chain	H-bonds; disulfide S-S bonds (covalent); hydrophobic interactions; ionic bonds	Urea, heat, heavy metals, organic solvents
4 degree	Quaternary structure	Assembly of 2 or more polypeptide subunits	Same as tertiary + disulfide bridges between chains	Same agents as tertiary structure

Peptide Bond Properties

- Formed by: R₁-COOH + H₂N-R₂ gives R₁-CO-NH-R₂ + H₂O (condensation reaction)
- Has PARTIAL double-bond character (resonance between N lone pair and C=O) = PLANAR and RIGID
- C-N peptide bond length: 1.32 Å — shorter than normal C-N single bond (1.47 Å)
- Mainly in TRANS configuration (R groups on opposite sides) — less steric strain
- N-terminus (free -NH₂) at LEFT by convention. C-terminus (free -COOH) at RIGHT

Tests for Proteins and Amino Acids

Test	Reagent	Colour Result	Detects	Conditions and Notes
Biuret test	CuSO ₄ + NaOH (aqueous)	Violet or purple colour	Peptide bonds — needs 2 or more peptide bonds	Single amino acids = NO reaction. Dipeptides = very faint. Proteins = deep violet. Cu ²⁺ coordinates with N atoms of peptide backbone.
Ninhydrin test	Ninhydrin solution	Purple (Ruhemann's purple)	ALL free alpha-amino acids	Proline gives YELLOW not purple (secondary amine — no free NH ₂). Very sensitive. Used in paper/TLC chromatography.
Xanthoproteic test	Conc. HNO ₃ ; then dilute NH ₃	Yellow colour turns orange with NH ₃	Aromatic amino acids: Phe, Tyr, Trp	Nitration of aromatic ring. Yellow nitro compound turns orange in alkali. Negative for proteins without aromatic amino acids.
Millon's test	Mercuric nitrate Hg(NO ₃) ₂ in HNO ₃	Red colour (precipitate or solution)	Tyrosine (phenolic -OH group)	Positive for ANY phenolic compound. Negative for proteins lacking tyrosine (e.g. gelatin).
Hopkins-Cole test	Glyoxylic acid + conc. H ₂ SO ₄	Purple ring at acid-protein interface	Tryptophan (indole ring)	Specific for Trp indole group.
Lead acetate test	Pb(OAc) ₂ + NaOH; boil	Black precipitate (PbS)	Cysteine and cystine (sulfur-containing amino acids)	Sulfur released as H ₂ S on heating; reacts with Pb ²⁺ to give black PbS precipitate.
Sakaguchi test	alpha-naphthol + NaOBr	Red colour	Arginine (guanidinium group)	Specific for arginine only.

Denaturation

Agent	Example	Effect on Structure	Reversible?
High heat	Boiling; cooking	Disrupts H-bonds and hydrophobic interactions — unfolding of 2/3/4 degree	Usually NO (e.g. cooking egg white)

Agent	Example	Effect on Structure	Reversible?
Extreme pH	Conc. HCl or NaOH	Alters ionisation of R groups — disrupts ionic bonds and H-bonds	Sometimes reversible if mild
Organic solvents	Ethanol, acetone	Disrupts hydrophobic interactions	Sometimes reversible after removal
Heavy metals	Pb ²⁺ , Hg ²⁺ , Ag ⁺	React with -SH groups (cysteine) — destroy disulfide bonds and enzyme active site	Generally irreversible
Urea or guanidine HCl	8 M urea	Disrupts H-bonds and hydrophobic interactions extensively	Sometimes reversible after careful removal

THINGS TO NOTE: THINGS TO NOTE: (1) Denaturation destroys 2, 3, and 4 degree structure but NEVER breaks peptide bonds (1 degree sequence = INTACT). (2) Denatured protein loses biological function — enzyme becomes inactive. (3) Essential amino acids (must come from diet, cannot be synthesised): Val, Leu, Ile, Phe, Trp, Met, Thr, Lys (and His for children). (4) Biuret: 2+ peptide bonds = violet. Single amino acid = no reaction. (5) Proteases/peptidases hydrolyse peptide bonds during digestion.

EXAM HACK: BIURET = 2+ peptide bonds = proteins (violet). NINHYDRIN = ALL alpha-amino acids (purple). XANTHOPROTEIC = yellow = aromatic AA. MILLON'S = red = tyrosine. LEAD ACETATE = black PbS = cysteine/cystine. Denaturation = SHAPE gone; sequence INTACT.

ALZHEIMER'S MODE: Proteins = necklaces of amino acid beads that fold into shapes (helix, sheet, 3D structure). Heat unfolds the necklace (denaturation) — like cooking an egg. You cannot uncook an egg. But the beads (amino acids) are still there. Biuret test: purple = protein confirmed.

CHAPTER 17 — MASTER QUICK REFERENCE

Boiling Point Order: Highest to Lowest (same MW)

Rank	Class	Example and BP	Why
1 Highest	Carboxylic acids RCOOH	Acetic acid 118 C	H-bond dimers: 2 H-bonds per pair
2	Alcohols ROH	Ethanol 78 C	Strong O-H...O H-bonds
3	Amines RNH ₂	n-propylamine 49 C	N-H...N H-bonds (weaker than O-H)
4	Aldehydes RCHO	Propanal 49 C	Polar C=O dipole only — no H-bond
5	Ketones RCOR	Acetone 56 C	Polar C=O dipole only
6	Esters RCOOR	Ethyl acetate 77 C	Polar but no H-bond donor
7	Ethers ROR	Diethyl ether 35 C	Weakly polar only
8	Alkyl halides RX	1-bromopropane 71 C	Polar but no H-bonding
9 Lowest	Alkanes RH	Butane -0.5 C	Only weak London dispersion forces

Acidity Order: Strongest to Weakest

Compound	pKa	Key Note
HCl	-7	Strong mineral acid — complete ionisation
CF ₃ COOH (trifluoroacetic)	0	3x F atoms = very powerful EWG
CCl ₃ COOH (trichloroacetic)	0.66	3x Cl = strong EWG
RCOOH (carboxylic acids)	4 to 5	Resonance-stabilised carboxylate
H ₂ CO ₃ (carbonic acid)	6.4	Dissolved CO ₂ in water
ArOH (phenol)	10	Phenoxide resonance-stabilised by benzene ring
H ₂ O	15.7	Reference
ROH (aliphatic alcohols)	16 to 18	No resonance stabilisation of alkoxide
Terminal alkynes RC triple CH	~25	sp carbon more electronegative

Compound	pKa	Key Note
RH (alkanes)	~50	Carbanion extremely unstable

Reactivity of Acid Derivatives

Rank	Derivative	LG Quality	Why	Hydrolysis Speed
1 MOST	Acid chloride (RCOCl)	Cl- (excellent)	Best LG; very electrophilic C=O	Fumes in air; reacts instantly with water
2	Anhydride (RCOOCOR)	RCOO- (moderate)	Decent LG	Reacts slowly with water
3	Ester (RCOOR')	RO- (poor)	Poor LG; needs activation	Needs acid/base + heat
4 LEAST	Amide (RCONH ₂)	NH ₂ - (terrible)	N lone pair stabilises C=O; terrible LG	Vigorous prolonged conditions

Oxidation Summary

Starting Material	Mild Oxidation (PCC or K ₂ Cr ₂ O ₇ controlled)	Strong Oxidation (KMnO ₄ or excess K ₂ Cr ₂ O ₇)
Primary alcohol (RCH ₂ OH)	Aldehyde (RCHO)	Carboxylic acid (RCOOH)
Secondary alcohol (R ₂ CHOH)	Ketone (RCOR)	Ketone — stops here
Tertiary alcohol (R ₃ COH)	No reaction	No reaction
Aldehyde (RCHO)	Carboxylic acid (RCOOH)	Carboxylic acid
Ketone (RCOR)	No reaction (mild)	Cleavage into 2 smaller carboxylic acids
Alkene (C=C)	Diol via syn dihydroxylation with OsO ₄	Cleavage into aldehydes or ketones then acids

ALL Colour Tests — Complete Master Reference

Test	Reagent	Colour Result	Detects	Key Note
Tollens (silver mirror)	AgNO ₃ / NH ₃	Silver mirror on glass	ALL aldehydes; reducing sugars	ALL aldehydes positive including benzaldehyde. Warm gently — never boil.
Fehling's	Fehling's A + B (blue)	Brick-red ppt Cu ₂ O	Aliphatic aldehydes; reducing sugars	Benzaldehyde = NEGATIVE. Positive for glucose, maltose, lactose, fructose.
Benedict's	Benedict's reagent (blue)	Orange-red ppt	Reducing sugars	Used in clinical glucose in urine testing.
2,4-DNPH (Brady's)	2,4-Dinitrophenyl hydrazine	Orange or yellow ppt	Aldehydes AND ketones (C=O)	MP of precipitate identifies specific compound.
Iodoform	I ₂ / NaOH	Yellow ppt (CHI ₃) + smell	CH ₃ CO- group; CH ₃ CHO; ethanol; CH ₃ CHOH-	Specific to methyl carbonyl or methyl carbinol.
Lucas test	ZnCl ₂ / conc. HCl	Cloudiness (turbidity)	Distinguishes 1, 2, 3 degree alcohols	3 degree = instant; 2 degree = 5 min; 1 degree = no rxn at RT.
Schiff's	Fuchsin + SO ₂	Pink or magenta	Aldehydes only	More sensitive than Tollens.
Molisch's	H ₂ SO ₄ + alpha-naphthol	Purple ring	ALL carbohydrates	Negative = not a carbohydrate.
Seliwanoff's	Resorcinol + conc. HCl	Cherry-red fast vs pink slow	Ketoses fast vs Aldoses slow	Fructose cherry-red in 2 min.
Bial's	Orcinol + HCl + FeCl ₃	Blue-green	Pentoses (5-carbon sugars)	Detects ribose in RNA.
Barfoed's	Copper acetate / acetic acid	Red ppt in 5 min	Monosaccharides vs disaccharides	Monosaccharides react in 5 min; disaccharides much longer.
Iodine / KI	I ₂ in KI solution	Blue-BLACK	Starch (amylose) only	Amylose = blue-black; Amylopectin = blue-purple; Glycogen = reddish-brown; Cellulose = none.
Biuret	CuSO ₄ / NaOH	Violet or purple	Proteins (2+ peptide bonds)	Single amino acids = no reaction.

Test	Reagent	Colour Result	Detects	Key Note
Ninhydrin	Ninhydrin solution	Purple (Ruhemann's purple)	ALL alpha-amino acids	Proline = yellow.
Xanthoproteic	Conc. HNO ₃ then NH ₃	Yellow then orange	Aromatic amino acids Phe Tyr Trp	Nitration of aromatic ring.
Millon's	Hg(NO ₃) ₂ / HNO ₃	Red colour	Tyrosine (phenolic -OH)	Any phenolic compound tests positive.
Hopkins-Cole	Glyoxylic acid + H ₂ SO ₄	Purple ring	Tryptophan (indole ring)	Specific for Trp.
Lead acetate	Pb(OAc) ₂ / NaOH; boil	Black ppt (PbS)	Cysteine and cystine (sulfur AA)	H ₂ S released reacts with Pb ²⁺ .
Carbylamine	CHCl ₃ + alcoholic KOH	Foul isocyanide smell	PRIMARY amines ONLY	2 degree and 3 degree = NO reaction.
Baeyer's	Dil. KMnO ₄ (purple)	Decolorised to brown	Unsaturation; aldehydes	Also reacts with aldehydes — not specific to alkenes alone.
Bromine water	Br ₂ / water (orange-brown)	Decolorised (turns clear)	Alkenes; alkynes; phenol; aldehydes	Immediate decolorisation = unsaturation or phenol.

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